簡宗宇

CHARLES JIAN

Application Architecture

About Me

1988/08/05

I am a technology enthusiast passionate about exploring innovations and tackling unique project challenges.
I maintain a positive attitude and excel in cross-team collaboration in high-pressure environments, empowering my team to achieve our goals.
Looking ahead, I aim to drive innovative solutions and architectures, including WebAssembly, containerization, and hybrid cloud technologies.

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Blogger   Git

Education

TAIPEI TECH 國立臺北科技大學 National Taipei University of Technology

Master Degree

Electrical Engineering Department

2010 - 2012

大同大學 TATUNG UNIVERSITY

Bachelor Degree

Department of Applied Mathematics

2006 - 2010

Skills

IDE/Infra

- Visual Studio
- Visual Studio Code
- SQL Server Management Studio
- Toad
- Source Tree
- Console Emulator
- PyCharm
- IIS
- Azure
- Redis
- Docker

Frontend

- HTML
- CSS/SASS/SCSS
- Javascript
- Jquery
- Vue
- Angular
- Bootstrap

Backend

- C#
- VB.Net
- C / C++
- Java
- (T/PL)-SQL
- Xml
- Xaml
- Yuml
- MarkDown
- Python

Issue Tracker

- TFS
- Jira
- Wekan
- Trello
- Sonarqube
- Git

DataBase

- MSSQL
- MySQL
- Oracle
- MongoDB

Version Control

- TFS
- Git
- SVN
- Perforce

Experience

IBM International Business Machines Corporation

Application Architecture *2019 – Now*

E.SUN Bank Core Transformation Project - Financial Cross-Border System

- As a System Architect and Designer for the core internal system at E.SUN Bank, the project aimed to enable cross-border application approvals among international staff. Key responsibilities included:Conducting interviews with primary users to gather requirements and clearly define the project scope.
- Facilitating consensus among stakeholders on system design early in the project.
- Designing specifications based on interview results and confirming alignment with client expectations.
- Monitoring progress during development and testing phases, reporting status to the project manager.
- Providing test plans during System Integration Testing (SIT) and User Acceptance Testing (UAT), facilitating issue tracking.
- Monitoring potential risks and reporting updates to the project manager.
- Tracking deliverables for each milestone and preparing for the handover process.
- Assisting the project manager in planning each milestone, allocating features and personnel, and controlling project quality and design.

The project successfully improved overall processes and operational models by 300%, enhanced efficiency by 80%, and increased maintenance efficiency by approximately 40%. I led a team of 13 members to complete this project.

Shin Kong Hospital - Human Resources System

In Shin Kong Hospital, I served as a System Designer and Developer, responsible for the scheduling system for resident doctors and the promotion system. I calculated each doctor's bonuses, allowances, and salary through relevant computation methods, ensuring accuracy and efficiency in the system.

WPG - BPaaS & LaaS

- In this project, I served as Development Leader, leading a team of six members. The system managed inventory using various Excel templates provided by partners. Before shipment, complex calculations were performed to generate results required by different vendors. Key responsibilities included:Coordinating with various aspects involved in the project to ensure seamless completion of tasks.
- Monitoring progress during development and testing phases, reporting status to the project manager.
- Training users on new tasks and workflows.
- Planning how to upload the system to Azure Cloud and AKS, integrating logs through ELK.
- Assisting with personnel configuration and allocation.

CTBC Bank - Navigator Core Project

As an Architect and System Analyst for CTBC Bank's major five-year, $800 million project, I was responsible for planning all systems into a middle-platform structure and migrating existing systems, classified into three areas: core systems integrated through Tata TCS, credit card business integrated through Worldline, and middle-platform systems handling currency exchange operations. This project improved internal service systems by 300% and tracked changes across core fields.

Fubon Bank - Foreign Debt/Self-operated Financial Market

At Fubon Bank, I served as PM, Architect, System Analyst, System Designer, and Developer, leading a team of five members in a comprehensive upgrade of the bank's architecture. Utilizing micro-frontends for integration, I optimized development speed by nearly 200%. Additionally, I was responsible for migrating AS400 RPG to open languages, integrating past and present domains while implementing a new permission control mechanism for process approvals. The system's service growth reached approximately 70%.

Trend Micro

Research and Development Engineer *2018 – 2019*

During my time at Trend Micro, I served as a Research and Development Engineer, primarily responsible for the Control Manager product. The core business of this product is to integrate various Trend Micro products, allowing for control and interfacing with other solutions through Control Manager. The product is divided into two versions: on-premise and SaaS, with my focus being on the installation of the on-premise version.

Given that this product has been in existence for nearly 20 years, I undertook a comprehensive overhaul of the automated installation process to gain a deeper understanding of all project components and streamline operations. Throughout the installation process, I was able to identify errors and version compatibility issues more effectively. As a result of my optimizations, security improved by approximately 40%, debugging efficiency increased by 70%, and compatibility enhanced by 30%.

The product underwent a significant General Availability (GA) release, during which I coordinated with the support team to clarify issues and collaboratively complete a major upgrade of the product while addressing problems in the legacy system. This product holds a market share of 98% in Japan and 43% in Taiwan.

Experience

Gamania Digital Entertainment (GASH)

System Architecture *2015 – 2018*

As a Senior Engineer at GASH, a subsidiary of Game Orange Group, I was responsible for the development and restructuring of payment transaction and membership systems. Initially, I took charge of the CPN advertising system, which included Facebook ads and various third-party integrations. Subsequently, I transitioned to a comprehensive overhaul of the membership system, encompassing the following technical architecture.

1. System Architecture: Reconstructed both front-end and back-end systems, including adjustments to DMZ network architecture, N-tier, N-Layer, load balancing, read-write separation, high availability (HA) architecture, full API integration, caching mechanisms, server validation mechanisms, and WebSocket asynchronous payment integration.
2. Database Management: Conducted database normalization adjustments (Always On) to ensure system stability
3. APP Development: Collaborated on the development of B2B and B2C applications providing 24/7 services.
4. Financial Management: Reorganized internal and external accounts, recovered all bad debts, and ensured the stability of all financial expenditures.

After undergoing two months of internal testing with significant speed adjustments that accelerated the original system by approximately three times, I received the MVP award monthly upon successful system launch and was promoted to System Architect. I led a team of around eight members responsible for maintaining and adjusting the payment system in collaboration with major banks for specifications and integration methods. The number of members grew from 1 million to over 20 million during this period. During the "Lineage M" event, we achieved a record of over 500 concurrent connections per second, with total connections reaching 6 million. Total revenue increased from 60-70 million to 300-400 million, with net profit growing to four times previous levels. Monitored through a dashboard, I successfully handled massive traffic consuming over 32 physical servers in approximately four hours. Additionally, the APP ranked first in downloads for two consecutive months.

THE SYSCOM GROUP

Junior Research and Development Engineer *2012 – 2015*

Prison Management Information System - Government Legal Department

The system is responsible for recording various behaviors and health parole records of inmates across Taiwan. The primary users include approximately 30 law enforcement personnel per prison, with a development team of over 100 members. The system architecture is based on a three-tier structure, utilizing WPF (MVVM) and WCF for front-end and back-end separation, while connecting to an MSSQL database. The system is ultimately deployed and updated online via OneClick, and the development and testing were completed within six months, followed by on-site drills in major prisons nationwide.

Responsibilities

1. Core Education: Responsible for the core educational functions of the system.
2. Course Scheduling and Seat Analysis: Conducted analysis for course scheduling and seat assignments.
3. Performance Scoring Calculation: Calculated inmate performance scores to support parole applications.
4. Scheduling Management: Managed the scheduling of law enforcement personnel.

Bus Arrival System - The Ministry of Transportation

The system evaluates the real-time location of buses and their distance to stops to calculate estimated arrival times, with a margin of error not exceeding 2 minutes. The system processes approximately 5 to 7 million messages daily and employs Solar technology to ensure asynchronous data transmission and communication.

Responsibilities

1. Station Planning: Inventorying the coordinates and distances of each bus stop.
2. Program Development: Developing system programs and integrating third-party information.
3. Real-Time Scheduling: Designing the overall architecture to support real-time data updates.
4. Reporting and Analysis: Generating various reports for statistical analysis.

Loan Management System Project - Taishin Bank

In the loan management system project at Taishin Bank, the internal system is utilized by approximately 30 users per branch, processing around 300,000 to 500,000 records daily. The main objective of the project was to convert a paper-based workflow into an electronic format, utilizing customized approval processes for verification. The system is developed using ASP.NET and VB.NET, with an MSSQL database, requiring customization in collaboration with system administrators.

Responsibilities

1. Template Provision: Provided downloadable templates for various required uploads.
2. Document Upload and Analysis: Uploaded various documents and analyzed customer application types to determine compliance with lending regulations.
3. Approval Verification: Ensured that approvals were verified by first-line supervisors; issued reminders via email for any returned applications requiring additional information.
4. Daily Reporting: Generated reports for managing various loan items and monitoring amount levels.

Experience

THE SYSCOM GROUP

Junior Research and Development Engineer 2012 – 2015

Core Accounting System - Yuanta Polaris Securities

In the core accounting system project at Yuanta Polaris Securities, Yuanta Securities integrated its legacy COBOL system with Polaris Securities' PowerShell system, ultimately evolving into a new system based on .NET(Windows) and Oracle(Linux) and Socket/C++(Radhat Linux). The system processes approximately 4 to 6 million transactions daily, with transaction amounts ranging from 50 to 80 million, and may exceed 100 million due to same-day cancellations mandated by regulations.

Responsibilities

1. Non-Functional Requirements: Prior to market opening, I uploaded data from 194 branches to the central custody center and completed all verifications and downloads within two hours for trading readiness.
2. Trade Matching: During trading, I applied a first-in-first-out principle using a linked list to facilitate real-time buy and sell transactions, updating stock prices and charts instantaneously.
3. Fee Calculation: Calculated various transaction fees for institutional and retail clients, including high-risk groups, processing all calculations within one second.
4. Test Coverage: Ensured test coverage exceeded 75%.
5. Stress Testing: Successfully conducted stress tests, processing 1.7 million records in just 17 seconds.

Postgraduate

2010 – 2012

Master's Thesis：An Approach for Floorplanning Massive Modules Based on CUDA Architecture

In my master's thesis, I independently improved upon existing algorithms for floorplanning massive modules, eliminating special cases and deriving mathematical proofs for new patterns. By leveraging GPU architecture for parallel computation, I achieved a significant increase in processing speed—at least seven times faster than the original referenced literature. Additionally, my approach reduced overall costs by at least 30% compared to benchmark data.

RFID Research

During my graduate studies, I worked on a National Science Council project involving RFID technology to help visually impaired individuals identify items through audio confirmations. I collaborated with a local association for the visually impaired to implement this solution.I also developed an RFID-based class sign-in system for my school, managing communication with RFID vendors and integrating the RS232 protocol for audio output. This project involved independent work with administrative units and database application development.

References

- 協助視障者辨別物品的RFID輔助系統
- 放鬆波蘭式用於大量模組之平面規劃
- 單元測試
- 高效能序列化的Google Protocol Buffers
- Microsoft Certified: Azure Fundamentals
- Microsoft Certified: Azure Developer Associate
- Microsoft Certified: Azure Security Engineer Associate
- Microsoft Certified: DevOps Engineer Expert
- Microsoft Certified: Azure Solutions Architect Expert